

backup data. Therefore, DVD rewritable (DVD-RW/+RW) and Blu-ray recordable/erasable (BD-RE) are preferred for regular backups. Moreover, the life span of an optical disc is more than 100 years!

Optical discs may be a better option for creating backups, but can suffer from physical damage if not handled carefully. Any scratch or cosmetic damage on the readable surface of the optical disc can corrupt its stored data. That's why another better option is to backup data on tapes. Tapes used to backup digital data are called Linear Tape-Open (LTO) Ultrium, and an LTO-8 tape cartridge can store up to 12TB of uncompressed data and 30TB of compressed data.

Tapes are good for making multiple copies of a backup because they can be used for duplication and replication. It also takes very little time to restore a backup using tapes and that's why many data centres still use tape drives in their backup repository. However, using tapes for backup at home is not recommended because the tape cartridges cost more than any optical disc. LTO Ultrium tapes have a better shelf life when compared to SDDs and can retain data up to 30 years.

Besides these storage media, backups can also be done on an online cloud storage platform like Google Drive. But for doing backups of large quantities of data on the cloud, high speed Internet with a good and stable connection is needed. It's also not a good idea to store large files on cloud storage drives, because during any network issues it will be difficult to restore the backup data. Also, cloud storage providers are not always reliable.

Popular FOSS tools for backup

Backup can be done on a reliable storage medium using good backup software. There are various FOSS tools available for the job, and choosing the right tool can help you carry out the backup process conveniently. A few of these FOSS tools are outlined below.

Bacula: This is a popular FOSS enterprise-level computer backup tool for heterogeneous networks, and can automate any backup that needs intervention from a systems administrator or computer operator. It supports various operating systems including Linux, UNIX, Windows and macOS, and a range of professional backup devices including tape libraries. Bacula can be configured and used with a command line console, GUI or Web interface. It was developed in January 2000 by Kern Sibbald and written in C and C++, which are very powerful computer languages. Bacula is open source and released under the AGPL version 3 license, but there is an exception to permit the linking with OpenSSL and distributing the Windows binaries. The firewall administration and network security of Bacula is easy to use because the TCP/IP client-server communication uses standard ports and services

instead of RPC for NFS, SMB, etc. It uses the CRAM-MD5 configurable client-server authentication and the GZIP/LZO client-side compression, which reduces the network bandwidth consumption. Bacula also uses transport layer security (TLS) network communication encryption, MD5/SHA file integrity verification, cyclic redundancy check (CRC) data block integrity verification, public key infrastructure (PKI) backup data encryption, network data management protocol (NDMP) enterprise version plugin, and supports cloud backup with some Amazon Simple Storage Service file storage services.

Amanda: This is another FOSS tool that can backup data from multiple computers in a network. It uses client-server architecture, and the server connects with the client to backup data at a scheduled time. It was developed at the University of Maryland in USA, and is written in C and Perl language. The commercial version, called the Amanda Enterprise Edition, was developed by Zmanda Inc. It can run on almost any UNIX operating system and also on Linux, Windows, macOS X and Solaris. Amanda can backup data on both tapes and disk drives. A special feature of Amanda is that it supports tape-spanning; so if the backup doesn't fit in one tape, it can be saved on multiple tapes by splitting the data. Another special feature is the intelligent scheduler, which can optimise the use of computing resources across many backup runs.

Back in Time: This FOSS tool is exclusively written for the Linux operating system. It was developed by a group of people, namely, Oprea Dan, Bart de Koning, Richard Bailey, Germar Reitze and Taylor Raack in 2008, and is written in Python. This backup tool is distributed along with many Linux distributions. Similar to Apple's Time Machine, this reliable backup tool uses rsync (a utility for synchronising files between various storage media) as a backend and hard links for file storage. This eliminates unnecessary usage of disk space, when there are identical snapshots of a file at different times. This hard link method also makes it easy to look at the snapshots of the system at different times and can help in the removal of any unnecessary snapshot. Back in Time also supports encryption of backups, and backups over SSH.

A disaster can happen at any time, and keeping data safe is the biggest challenge for all businesses. Backups can help recover the data after any disaster, but it is important to choose the right backup method and storage medium. These backups can be automated using a reliable tool, and choosing the right one can make the process much easier. The FOSS backup tools discussed here can be used to keep data safe. 

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